Exp. no: 09
Date: 10/3/26

Determination of Z parameter of two port networks

Aim:

To calculate and verify h parameters of a two port network by using Multisim Simulator

Apparatus required:

S.no	Components	Specification	Quantity
01	Power supply	-	1
02	Multimeter	-	1
03	Resistor	10K, 10K, 4.7K	
04	Bread board	-	1
05	Connecting wires	-	-
06	Multisim Simulator	-	-

Procedure:

a) Connect the circuit

b) Short the output port and excite input port with known voltage

c) input port is open circuited and output port is excited with same

d) Switch off the supply after taking the readings

Observation :

S.no	Output open Circuited ($I_2 = 0$)			$Z_{21} = V_2/I_1 (\Omega)$	$Z_{11} = V_1/I_1 (\Omega)$
	V_1	V_2	I_1		
1	5V	3.34V	2.30mA	1.54KΩ	2.17KΩ

$Z_{12} = \frac{V_1}{I_1} = \frac{3.34}{0.023}$ $\qquad Z_{22} = \frac{V_2}{I_2} = \frac{5}{0.0023}$

$Z_{12} \approx 1539\,\Omega$ $\qquad Z_{22} \approx 2174\,\Omega$

S.no	Input open Circuited ($I_1 = 0$)			$Z_{12} = V_1/I_2 (\Omega)$	$Z_{22} = V_2/I_2 (\Omega)$
	V_2	V_1	I_2		
1	5V	3.34V	2.30mA	1.54KΩ	2.17KΩ

$Z_{12} = \frac{V_1}{I_2} = \frac{3.34}{0.023}$ $\qquad Z_{22} = \frac{V_2}{I_2} = \frac{5}{0.0023}$

$Z_{12} \approx 1539\,\Omega$ $\qquad Z_{22} \approx 2174\,\Omega$

O/P verified 11/8/24

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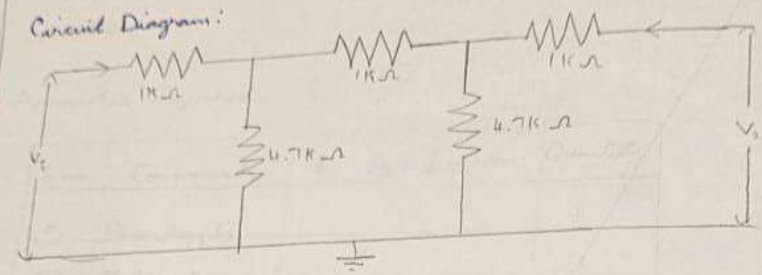
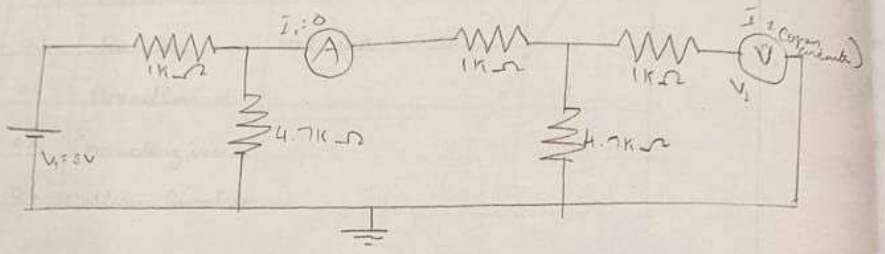
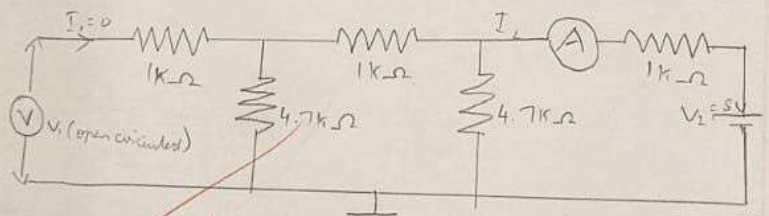
Sl.no	Description	Marks allocated	Marks alloted
01	Experimental setup	20	20
02	observation	20	20
03	Calculation	20	20
04	Viva	20	20
05	Record	20	20
06	Total	100	100

100

11/8/26

Result:

The Z-parameters of the two ports networks has been calculated and verified by using multisim simulator

Circuit Diagram: